

system's file set and compares it to previous logs of the file set.

Approaches to detecting intrusion

Signature based IDS: This focuses on searching for 'signature' patterns, or an identity of an intrusion, or a specific intrusion event. Signature-based IDS is only as good as how up-to-date its database is at that given moment. But it is very easy to bypass it by making tiny changes to the code. As the database increases, the processing load increases and so the system takes time to analyse each connection and verify it.

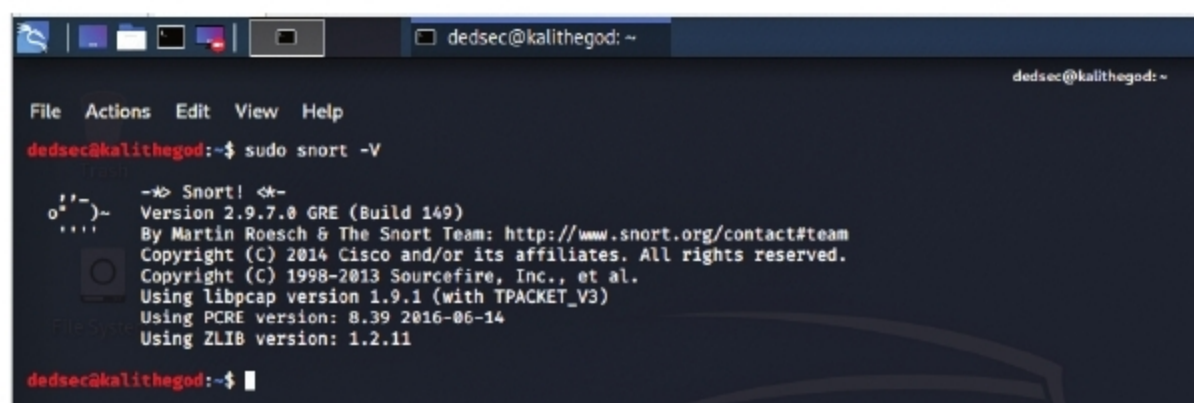
Anomaly based IDS: This uses machine learning to detect intrusions by comparing trustworthy models with new models. As a result, strange looking anomalies are flagged. However, the problem is that even a legitimate model can be flagged, depending on the network's response. Anomaly-based IDS is good for determining when someone is probing a network prior to an attack taking place. These probes create signals in the network that the anomaly based IDS will pick up.

Challenges of managing an IDS

1. False positives are a major problem with an IDS. This refers to the generation of alerts even when there is no real problem or any issue.
2. Staffing means that an IDS cannot work as a one-for-all system — it changes from enterprise to enterprise.
3. There are chances that an IDS may miss a legitimate risk.

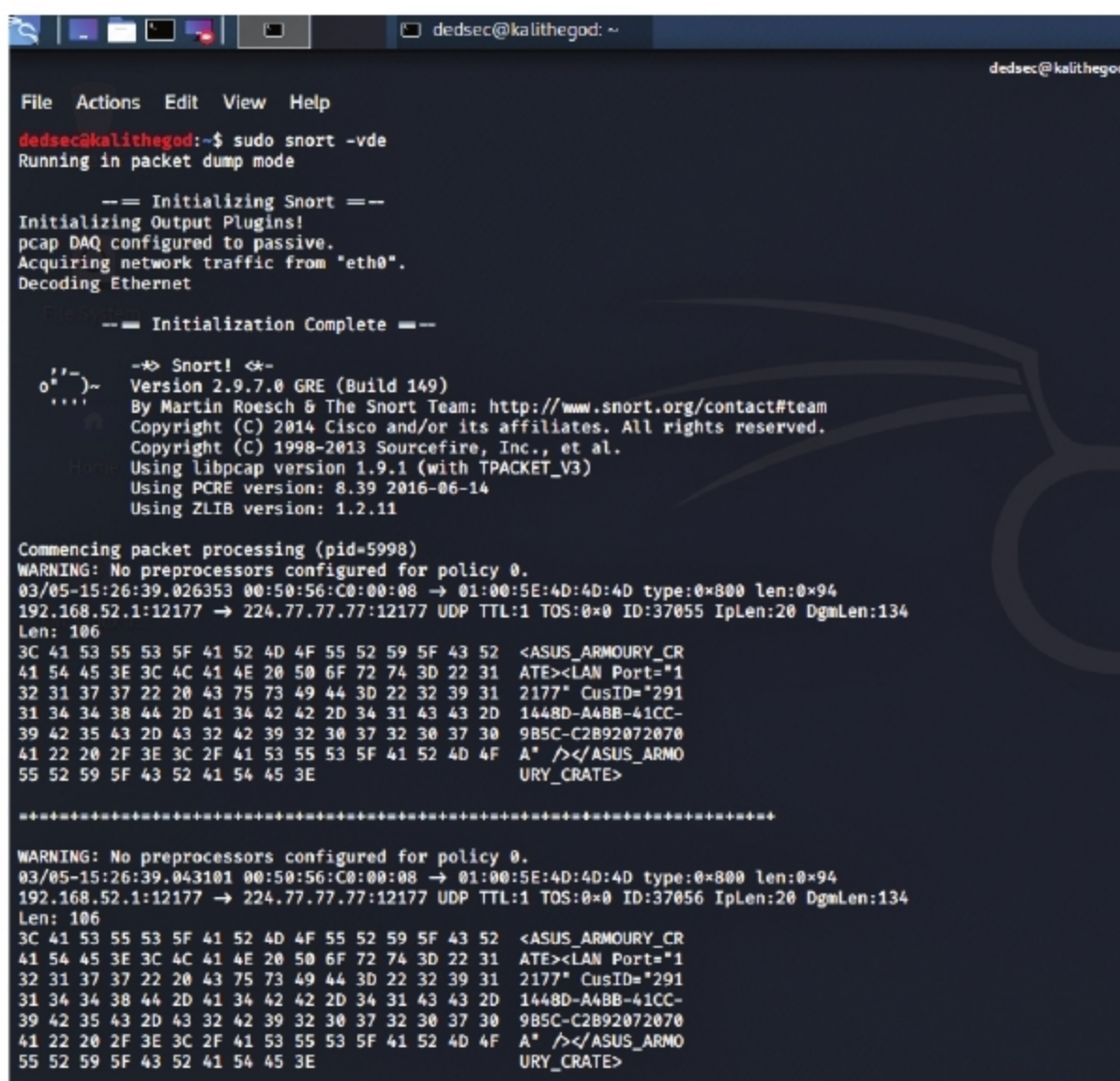
Snort

Snort is a free and open source network intrusion prevention and detection system. It uses a rule-based language combining signature, protocol and anomaly inspection methods to detect any kind of malicious activity. Snort is also capable of performing real-time traffic analysis and packet logging on IP networks. It was developed in 1999 but



```
dedsec@kalithead:~$ sudo snort -V
--> Snort! <--
Version 2.9.7.0 GRE (Build 149)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.9.1 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11
dedsec@kalithead:~$
```

Figure 1: Verifying the installation



```
dedsec@kalithead:~$ sudo snort -vde
Running in packet dump mode

-- Initializing Snort --
Initializing Output Plugins!
pcap DAQ configured to passive.
Acquiring network traffic from "eth0".
Decoding Ethernet

-- Initialization Complete --

--> Snort! <--
Version 2.9.7.0 GRE (Build 149)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.9.1 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11

Commencing packet processing (pid=5998)
WARNING: No preprocessors configured for policy 0.
03/05-15:26:39.026353 00:50:56:C0:00:08 -> 01:00:5E:4D:4D:4D type:0x800 len:0x94
192.168.52.1:12177 -> 224.77.77.77:12177 UDP TTL:1 TOS:0x0 ID:37055 IpLen:20 DgmLen:134
Len: 106
3C 41 53 55 53 5F 41 52 4D 4F 55 52 59 5F 43 52 <ASUS_ARMOURY_CR
41 54 45 3E 3C 4C 41 4E 20 50 6F 72 74 3D 22 31 ATE><LAN Port="1
32 31 37 37 22 20 43 75 73 49 44 3D 22 32 39 31 2177" CusID="291
31 34 34 38 44 2D 41 34 42 42 2D 34 31 43 43 2D 1448D-A48B-41CC-
39 42 35 43 2D 43 32 42 39 32 30 37 32 30 37 30 9B5C-C2B92072070
41 22 20 2F 3E 3C 2F 41 53 55 53 5F 41 52 4D 4F A" /></ASUS_ARMO
55 52 59 5F 43 52 41 54 45 3E URY_CRATE>

*****

WARNING: No preprocessors configured for policy 0.
03/05-15:26:39.043101 00:50:56:C0:00:08 -> 01:00:5E:4D:4D:4D type:0x800 len:0x94
192.168.52.1:12177 -> 224.77.77.77:12177 UDP TTL:1 TOS:0x0 ID:37056 IpLen:20 DgmLen:134
Len: 106
3C 41 53 55 53 5F 41 52 4D 4F 55 52 59 5F 43 52 <ASUS_ARMOURY_CR
41 54 45 3E 3C 4C 41 4E 20 50 6F 72 74 3D 22 31 ATE><LAN Port="1
32 31 37 37 22 20 43 75 73 49 44 3D 22 32 39 31 2177" CusID="291
31 34 34 38 44 2D 41 34 42 42 2D 34 31 43 43 2D 1448D-A48B-41CC-
39 42 35 43 2D 43 32 42 39 32 30 37 32 30 37 30 9B5C-C2B92072070
41 22 20 2F 3E 3C 2F 41 53 55 53 5F 41 52 4D 4F A" /></ASUS_ARMO
55 52 59 5F 43 52 41 54 45 3E URY_CRATE>
```

Figure 2: Sniffer output1

it went on to become so popular that it was purchased by Cisco in 2014.

Anyone, a novice or an experienced person, who wants to head out into the world of security must be familiar with Kali Linux. This is a Debian based distribution, which has been specifically made keeping in mind network analysts and penetration testers. A Linux distribution is a bundle which contains the Linux kernel, a collection of core utilities and applications, and some default settings. Kali comes pre-installed with a palette

of tools to choose from, making it the Swiss knife for hackers. We will now use Snort on Kali.

Installing Snort

Let us install Snort from the repositories. You can first directly try running it from the terminal, by using the following command — if your machine is properly configured with the repositories, then Snort will be installed.

```
kali > sudo apt-get install snort
```